1.	What is the range of well known port numbers?	0 to 1023
2.	What is the range of registered port numbers?	1024 to 49151
3.	What is the valid range of VLAN numbers	1 to 4094
4.	FTP	TCP port 20 TCP port 21
5.	Telnet	TCP port 23
6.	SMTP	TCP port 25
7.	HTTP	TCP port 80
8.	POP3	TCP port 110
9.	HTTPS (Secure HTTP)	TCP port 443
10.	TFTP	UDP port 69
11.	RADIUS Authentication Protocol	UDP port 1812
12.	DNS	TCP and UDP port 53
13.	SNMP	Agents: UDP port 161 NMS: UDP port 162
14.	Chef	TCP 10002
15.	What are the levels of the Syslog messages?	0 - Emergency 1 - Alert

2 - Critical
3 - Error
4 - Warning
5 - Notice
6 - Informational
7 - Debugging

Every Awesome Cisco Engineer Will Need Information Daily

16. **Puppet**

TCP 8140

17. **What is 172 in binary**

10101100

Radix	2	2	2	2	2	2	2	2
Position in Number	7	6	5	4	3	2	1	0
Calculate	(2^7)	(2^6)	(2^5)	(2^4)	(2^3)	(2^2)	(2^1)	(2^0)
Position value	128	64	32	16	8	4	2	1

18. **What is 192 in binary**

11000000

Radix	2	2	2	2	2	2	2	2
Position in Number	7	6	5	4	3	2	1	0
Calculate	(2^7)	(2^6)	(2^5)	(2^4)	(2^3)	(2^2)	(2^1)	(2^0)
Position value	128	64	32	16	8	4	2	1

19. **What is 168 in binary**

10101000

Radix	2	2	2	2	2	2	2	2
Position in Number	7	6	5	4	3	2	1	0
Calculate	(2^7)	(2^6)	(2^5)	(2^4)	(2^3)	(2^2)	(2^1)	(2^0)
Position value	128	64	32	16	8	4	2	1

20. **What is the administrative distance of a Directly Connected route?**

0

Route Source	Administrative Distance
Directly connected	0
Static route	1
EIGRP summary route	5
External BGP	20
Internal EIGRP	90
OSPF	110
IS-IS	115
RIP	120
External EIGRP	170
Internal BGP	200

21. **What is the administrative distance of a static route?**

1

Route Source	Administrative Distance
Directly connected	0
Static route	1
EIGRP summary route	5
External BGP	20
Internal EIGRP	90
OSPF	110
IS-IS	115
RIP	120
External EIGRP	170
Internal BGP	200

22. What is the administrative distance of EIGRP

90

Distance Vector	
[redacted]	EIGRP
[redacted]	EIGRP for IPv6

23. What is the administrative distance of OSPF

110

Link State	
OSPFv2	
OSPFv3	ipv6

24. What is the administrative distance of IS-IS

115

Link State	
[redacted]	IS-IS
[redacted]	IS-IS for IPv6

25. What is the administrative distance of RIP

120

Link State	
[redacted]	IS-IS
[redacted]	IS-IS for IPv6

26. What is the administrative distance of an unknown route

255

Route Source	Administrative Distance
Directly connected	0
Static route	1
EIGRP summary route	5
External BGP	20
Internal EIGRP	90
OSPF	110
IS-IS	115
RIP	120
External EIGRP	170
Internal BGP	200

27. What are the 2 modes for PAgP

Auto & Desirable

SwitchB / SwitchA	off	auto	desirable	passive	active	on
off	NO	NO	NO	NO	NO	NO
auto	NO	NO	PAgP	NO	NO	NO
desirable	NO	PAgP	PAgP	NO	NO	NO
passive	NO	NO	NO	NO	LACP	NO
active	NO	NO	NO	LACP	LACP	NO
on	NO	NO	NO	NO	NO	ON

28. What are the 2 modes for LACP

Active & Passive

SwitchB / SwitchA	off	auto	desirable	passive	active	on
off	NO	NO	NO	NO	NO	NO
auto	NO	NO	PAgP	NO	NO	NO
desirable	NO	PAgP	PAgP	NO	NO	NO
passive	NO	NO	NO	NO	LACP	NO
active	NO	NO	NO	LACP	LACP	NO
on	NO	NO	NO	NO	NO	ON

29. What is the mode that enables Ether-Channel

On & On

SwitchB / SwitchA	off	auto	desirable	passive	active	on
off	NO	NO	NO	NO	NO	NO
auto	NO	NO	PAgP	NO	NO	NO
desirable	NO	PAgP	PAgP	NO	NO	NO
passive	NO	NO	NO	NO	LACP	NO
active	NO	NO	NO	LACP	LACP	NO
on	NO	NO	NO	NO	NO	ON

30. What IEEE standard supports native VLANs & trunking?

802.1Q

31. What IEEE standard supports regular STP?

802.1D

32. What IEEE standard supports rapid STP?

802.1W

33. What IEEE standard supports LACP?

802.3ad

34. NTP

UDP port 123

35. What is the IEEE standard for LLDP?

802.1AB

36. Which frequency band(s) does the 802.11ac (Wi-Fi 5) standard operate on?

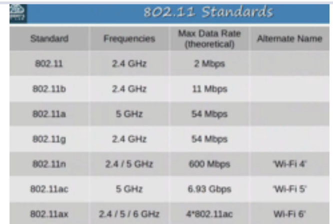
5GHz

802.11 Standards

Standard	Frequencies	Max Data Rate (theoretical)	Alternate Name
802.11	2.4 GHz	2 Mbps	
802.11b	2.4 GHz	11 Mbps	
802.11a	5 GHz	54 Mbps	
802.11g	2.4 GHz	54 Mbps	
802.11n	2.4 / 5 GHz	600 Mbps	'Wi-Fi 4'
802.11ac	5 GHz	6.93 Gbps	'Wi-Fi 5'
802.11ax	2.4 / 5 / 6 GHz	4*802.11ac	Wi-Fi 6'

37. Which frequency band(s) does the 802.11ax (Wi-Fi 6) standard operate on?

2.4GHz, 5GHz and 6GHz

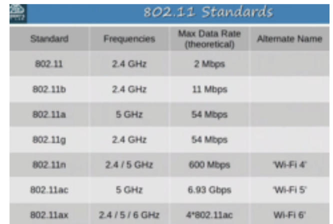


802.11 Standards

Standard	Frequencies	Max Data Rate (theoretical)	Alternate Name
802.11	2.4 GHz	2 Mbps	
802.11b	2.4 GHz	11 Mbps	
802.11a	5 GHz	54 Mbps	
802.11g	2.4 GHz	54 Mbps	
802.11n	2.4 / 5 GHz	600 Mbps	'Wi-Fi 4'
802.11ac	5 GHz	6.93 Gbps	'Wi-Fi 5'
802.11ax	2.4 / 5 / 6 GHz	4*802.11ac	Wi-Fi 6'

38. **Which frequency band(s) does the 802.11g standard operate on?**

2.4GHz

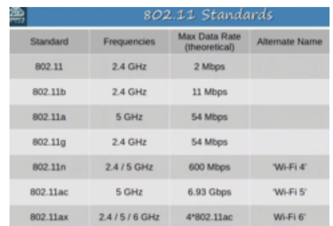


802.11 Standards

Standard	Frequencies	Max Data Rate (theoretical)	Alternate Name
802.11	2.4 GHz	2 Mbps	
802.11b	2.4 GHz	11 Mbps	
802.11a	5 GHz	54 Mbps	
802.11g	2.4 GHz	54 Mbps	
802.11n	2.4 / 5 GHz	600 Mbps	'Wi-Fi 4'
802.11ac	5 GHz	6.93 Gbps	'Wi-Fi 5'
802.11ax	2.4 / 5 / 6 GHz	4*802.11ac	Wi-Fi 6'

39. **Which frequency band(s) does the 802.11a standard operate on?**

5GHz

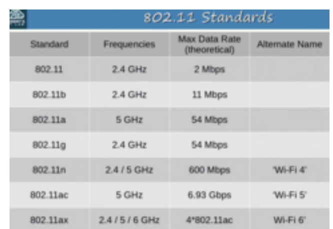


802.11 Standards

Standard	Frequencies	Max Data Rate (theoretical)	Alternate Name
802.11	2.4 GHz	2 Mbps	
802.11b	2.4 GHz	11 Mbps	
802.11a	5 GHz	54 Mbps	
802.11g	2.4 GHz	54 Mbps	
802.11n	2.4 / 5 GHz	600 Mbps	'Wi-Fi 4'
802.11ac	5 GHz	6.93 Gbps	'Wi-Fi 5'
802.11ax	2.4 / 5 / 6 GHz	4*802.11ac	Wi-Fi 6'

40. **Which frequency band(s) does the 802.11n (Wi-Fi 4) standard operate on?**

2.4GHz and 5GHz

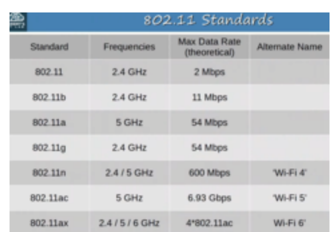


802.11 Standards

Standard	Frequencies	Max Data Rate (theoretical)	Alternate Name
802.11	2.4 GHz	2 Mbps	
802.11b	2.4 GHz	11 Mbps	
802.11a	5 GHz	54 Mbps	
802.11g	2.4 GHz	54 Mbps	
802.11n	2.4 / 5 GHz	600 Mbps	'Wi-Fi 4'
802.11ac	5 GHz	6.93 Gbps	'Wi-Fi 5'
802.11ax	2.4 / 5 / 6 GHz	4*802.11ac	Wi-Fi 6'

41. **Which frequency band(s) does the 802.11b standard operate on?**

2.4GHz

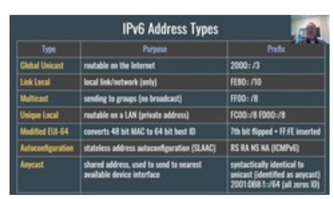


802.11 Standards

Standard	Frequencies	Max Data Rate (theoretical)	Alternate Name
802.11	2.4 GHz	2 Mbps	
802.11b	2.4 GHz	11 Mbps	
802.11a	5 GHz	54 Mbps	
802.11g	2.4 GHz	54 Mbps	
802.11n	2.4 / 5 GHz	600 Mbps	'Wi-Fi 4'
802.11ac	5 GHz	6.93 Gbps	'Wi-Fi 5'
802.11ax	2.4 / 5 / 6 GHz	4*802.11ac	Wi-Fi 6'

42. **What's the IPv6 multicast address space?**

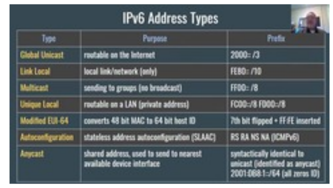
FF00::/8



IPv6 Address Types

Type	Purpose	Prefix
Global Unicast	routable on the Internet	2000:: /3
Link Local	local link/network (only)	FE80:: /10
Multicast	sending to groups (no broadcast)	FF00:: /8
Unique Local	routable on a LAN (private address)	FC00::/8 FD00::/8
Modified EUI-64	converts 48 bit MAC to 64 bit host ID	7th bit flipped + FF:FE inserted
Autoconfiguration	stateless address autoconfiguration (SLAAC)	RS RA NS NA (ICMPv6)
Anycast	shared address, used to send to nearest available device interface	syntactically identical to unicast (identified as anycast) 2001:DB8:1::/64 (all zeros ID)

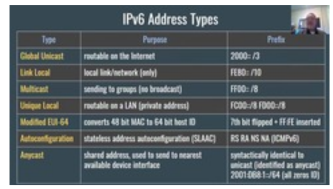
43.	What's the IPv6 unique local address space?	FC00::/7



IPv6 Address Types

Type	Purpose	Prefix
Global Unicast	routable on the Internet	2000:: /3
Link Local	local link/network (only)	FE80:: /10
Multicast	sending to groups (no broadcast)	FF00:: /8
Unique Local	routable on a LAN (private address)	FC00::/8 FD00::/8
Modified EUI-64	converts 48 bit MAC to 64 bit host ID	7th bit flipped + FF:FE inserted
Autoconfiguration	stateless address autoconfiguration (SLAAC)	RS RA NS NA (ICMPv6)
Anycast	shared address, used to send to nearest available device interface	syntactically identical to unicast (identified as anycast) 2001:DB8:1::/64 (all zeros ID)

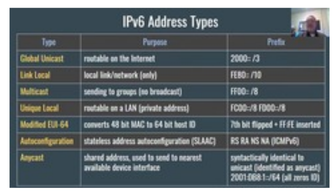
44.	What's the global unicast IPv6 address space?	2000::/3



IPv6 Address Types

Type	Purpose	Prefix
Global Unicast	routable on the Internet	2000:: /3
Link Local	local link/network (only)	FE80:: /10
Multicast	sending to groups (no broadcast)	FF00:: /8
Unique Local	routable on a LAN (private address)	FC00::/8 FD00::/8
Modified EUI-64	converts 48 bit MAC to 64 bit host ID	7th bit flipped + FF:FE inserted
Autoconfiguration	stateless address autoconfiguration (SLAAC)	RS RA NS NA (ICMPv6)
Anycast	shared address, used to send to nearest available device interface	syntactically identical to unicast (identified as anycast) 2001:DB8:1::/64 (all zeros ID)

45.	What's the IPv6 link local address prefix?	FE80::/64



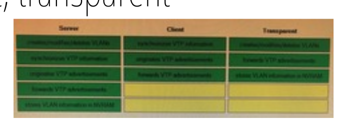
IPv6 Address Types

Type	Purpose	Prefix
Global Unicast	routable on the Internet	2000:: /3
Link Local	local link/network (only)	FE80:: /10
Multicast	sending to groups (no broadcast)	FF00:: /8
Unique Local	routable on a LAN (private address)	FC00::/8 FD00::/8
Modified EUI-64	converts 48 bit MAC to 64 bit host ID	7th bit flipped + FF:FE inserted
Autoconfiguration	stateless address autoconfiguration (SLAAC)	RS RA NS NA (ICMPv6)
Anycast	shared address, used to send to nearest available device interface	syntactically identical to unicast (identified as anycast) 2001:DB8:1::/64 (all zeros ID)

46.	What's the IPv6 loopback address	::1/128
47.	How is an EUI-64 address generated?	Split MAC address, insert FFFE in the middle and invert 7th bit.
48.	What's the default STP bridge priority value?	32768
49.	What are the modes for DTP	access, trunk, dynamic auto, dynamic desirable

Administrative mode	access	dynamic auto	dynamic desirable	trunk
access	access	access	access	misconfig
dynamic auto	access	access	trunk	trunk
dynamic desirable	access	trunk	trunk	trunk
trunk	misconfig	trunk	trunk	trunk

50.	What are the modes for VTP	server, client, transparent



Server	Client	Transparent
creates/modifies/deletes VLANs	synchronizes VTP information	creates/modifies/deletes VLANs
synchronizes VTP information	originates VTP advertisements	forwards VTP advertisements
originates VTP advertisements	forwards VTP advertisements	stores VLAN information in NVRAM
forwards VTP advertisements		
stores VLAN information in NVRAM		

51.	What plane is in the Southbound Interface	Data/Forwarding Plane
52.		Control Plane

What plane is in the Northbound Interface

53. Which FHRP uses a virtual MAC address of 0000.5e00.01xx — VRRP

FHRP	Terminology	Multicast IP	Virtual MAC	Cisco proprietary?
HSRP	Active/Standby	v1: 224.0.0.2 v2: 224.0.0.102	v1: 0000.0c07.acXX v2: 0000.0c9f.fXXX	Yes
VRRP	Master/Backup	224.0.0.18	0000.5e00.01XX	No
GLBP	AVG / AVF	224.0.0.102	0007.b400.XXYY	Yes

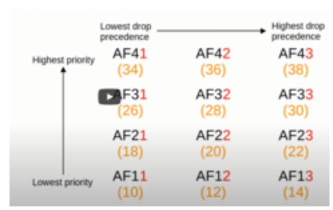
54. Which FHRP uses a virtual MAC address of 0007.b400.xxyy — GLBP

FHRP	Terminology	Multicast IP	Virtual MAC	Cisco proprietary?
HSRP	Active/Standby	v1: 224.0.0.2 v2: 224.0.0.102	v1: 0000.0c07.acXX v2: 0000.0c9f.fXXX	Yes
VRRP	Master/Backup	224.0.0.18	0000.5e00.01XX	No
GLBP	AVG / AVF	224.0.0.102	0007.b400.XXYY	Yes

55. Which FHRP uses virtual MAC addresses of 0000.0C07.ACxx & 0000.0C9F.Fxxx — HSRP

FHRP	Terminology	Multicast IP	Virtual MAC	Cisco proprietary?
HSRP	Active/Standby	v1: 224.0.0.2 v2: 224.0.0.102	v1: 0000.0c07.acXX v2: 0000.0c9f.fXXX	Yes
VRRP	Master/Backup	224.0.0.18	0000.5e00.01XX	No
GLBP	AVG / AVF	224.0.0.102	0007.b400.XXYY	Yes

56. What is the DSCP marking value for voice traffic? — 46 - EF



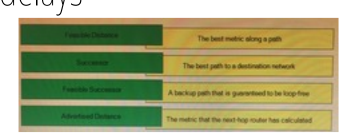
57. What is the recommended COS value for voice traffic? — 5

58. what wireless QOS profile is good for video traffic? — Gold

59. what wireless QOS profile is good for voice traffic? — Platinum

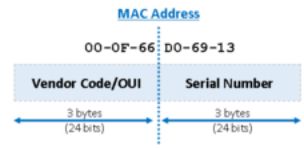
60. what is the default QOS profile that provides best-effort delivery? — Silver

61.	What is the range of standard ACLs?	1-99 and 1300-1999
62.	in NAT, which IP is the host from the perceptive of the local network?	Inside Local
63.	in NAT, which IP is the host from the perceptive of the outside network & hosts?	Inside Global
64.	What encoded languages are used for REST APIs?	JSON & XML
65.	What HTTP method creates a new variable	POST
66.	What HTTP method retrieves (read) the valuable of a variable	GET
67.	What HTTP method changes (update) the valuable of a variable	PUT /PATCH
68.	What HTTP method deletes a variable	DELETE
69.	What is the default CDP message runtime & hold time?	60/180 seconds
70.	What is the default LLDP message runtime & hold time?	30/120 seconds
71.	Which configuration tool uses SSH to connect to managed devices and uses Playbooks to manage tasks that's written in YAML?	Ansible

72.	Which configuration tool is written in Ruby to pull device configurations by using Manifest?	Puppet
73.	Which configuration tool is written in Ruby to pull device configurations by using Cookbooks?	Chef
74.	What are the OSPF neighbor states?	1. Down 2. Init 3. 2-way 4. Exstart 5. Exchange 6. Loading 7. Full IT ELF Init Two-way Extart/Exchange Loading Full
75.	What is the default reference bandwidth & cost in OSPF	100 mbps/10
76.	What wireless QOS profile provides the lowest bandwidth & is typically used for guest services?	Bronze
77.	What are the 2 factors that calculate the metric of EIGRP?	Lowest bandwidth The sum of delays 
78.	What's the purpose of a floating static route?	To provide a backup route (redundancy) to multiple interfaces/networks on a router

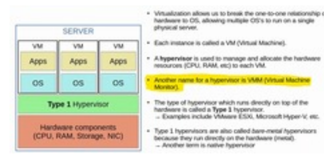
79. How many octets of a MAC address represent the OUI?

3



80. What is another name for a hypervisor?

VMM (Virtual Machine Monitor)



81. A company licenses an office suite including email service, that is delivered to the end user through a web browser. What type of service does this represent?

SaaS (Software as a Service)

82. What factors about FlexConnect ACLs are true?

They are supported on the native VLAN
They Cannot be configured with a per-rule direction
They are applied per AP & per VLAN

83. What are the functions performed by a WLC in a split-MAC deployment

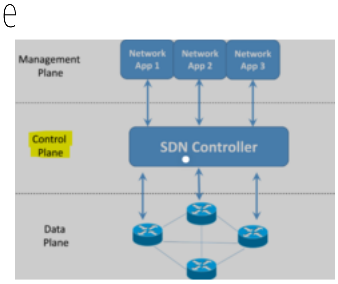
Authentication
Resource reservation

84. What encryption protocol is used for WPA2 & WPA3?

AES

85. Which plane is centralized on an SDN network?

Control Plane



86.

OnePK

Which southbound API is Cisco proprietary?

OnePK	is a Cisco proprietary API
OpenFlow	uses an imperative SDN model
OpFlex	uses a declarative SDN model
NETCONF	uses XML and RPCs to configure network devices

87. What is the default port security violation mode that shuts the port down when an unauthorized MAC address is detected?

Shutdown

88. Which port security violation mode discards traffic from unauthorized MAC addresses, and increments the violation counter by 1 for each unauthorized frame that's entered?

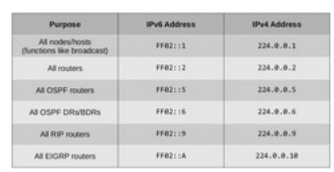
Restrict

89. Which port security violation mode that only discards traffic from unauthorized MAC addresses?

Protect

90. What's the IPv6 multicast address that routes to all nodes/hosts in the subnet?

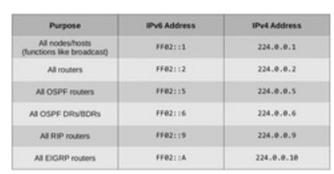
FF02::1



Purpose	IPv6 Address	IPv4 Address
All nodes/hosts (functions like broadcast)	FF02::1	224.0.0.1
All routers	FF02::2	224.0.0.2
All OSPF routers	FF02::5	224.0.0.5
All OSPF DRs/BDRs	FF02::6	224.0.0.6
All RIP routers	FF02::9	224.0.0.9
All EIGRP routers	FF02::A	224.0.0.10

91. What's the IPv6 multicast address that routes to all routers in the subnet?

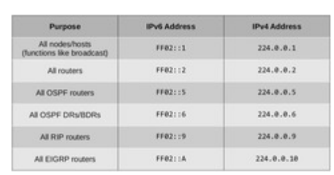
FF02::2



Purpose	IPv6 Address	IPv4 Address
All nodes/hosts (functions like broadcast)	FF02::1	224.0.0.1
All routers	FF02::2	224.0.0.2
All OSPF routers	FF02::5	224.0.0.5
All OSPF DRs/BDRs	FF02::6	224.0.0.6
All RIP routers	FF02::9	224.0.0.9
All EIGRP routers	FF02::A	224.0.0.10

92. What's the IPv6 link-local multicast address space?

FF02::/16



Purpose	IPv6 Address	IPv4 Address
All nodes/hosts (functions like broadcast)	FF02::1	224.0.0.1
All routers	FF02::2	224.0.0.2
All OSPF routers	FF02::5	224.0.0.5
All OSPF DRs/BDRs	FF02::6	224.0.0.6
All RIP routers	FF02::9	224.0.0.9
All EIGRP routers	FF02::A	224.0.0.10

93.

What is the default OSPF network type and interfaces?

Broadcast
Ethernet
FDDI

Broadcast	Point-to-point
Default on **Ethernet**, **FDDI** interfaces	Default on **HDLC**, **PPP** (serial) interfaces
DR/DBR elected	No DR/BDR
Neighbors dynamically discovered	Neighbors dynamically discovered
Default timers: Hello 10, Dead 40	Default timers: Hello 10, Dead 40

(**Non-broadcast** network type default timers = Hello 30, Dead 120)

94. What is the OSPF network type and interfaces that use serial cables, and does not participate in DR/BDR elections?

Point-to-Point
PPP
HDLC (default)

Broadcast	Point-to-point
Default on **Ethernet**, **FDDI** interfaces	Default on **HDLC**, **PPP** (serial) interfaces
DR/DBR elected	No DR/BDR
Neighbors dynamically discovered	Neighbors dynamically discovered
Default timers: Hello 10, Dead 40	Default timers: Hello 10, Dead 40

(**Non-broadcast** network type default timers = Hello 30, Dead 120)

95. What is the OSPF network type and interfaces that has a hello/dead timer of 30/120 seconds?

Non-Broadcast
Frame Relay
X.25

Broadcast	Point-to-point
Default on **Ethernet**, **FDDI** interfaces	Default on **HDLC**, **PPP** (serial) interfaces
DR/DBR elected	No DR/BDR
Neighbors dynamically discovered	Neighbors dynamically discovered
Default timers: Hello 10, Dead 40	Default timers: Hello 10, Dead 40

(**Non-broadcast** network type default timers = Hello 30, Dead 120)